

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME:** MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct

I **Gayathri N P (192421362)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls.
 - Box A contains 6 red, 4 blue, and 2 green balls.
 - Box B contains 5 red, 7 blue, and 3 green balls.
 - Box C contains 8 red, 2 blue, and 5 green balls.

One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue.

What is the probability that the selected box was Box C?

ANSWER :

Concept

This problem can be solved using **Bayes' Theorem**, which is used to find the probability of an event based on given evidence. Since a blue ball has been drawn, we need to find the probability that it came from **Box C**.

Given

- Box A = 6 Red, 4 Blue, 2 Green (Total = 12)
- Box B = 5 Red, 7 Blue, 3 Green (Total = 15)
- Box C = 8 Red, 2 Blue, 5 Green (Total = 15)

Selection probabilities:

- $P(A) = 1/4$
- $P(B) = 1/2$
- $P(C) = 1/4$

Probability of selecting a blue ball:

- $P(\text{Blue}|A) = 4/12 = 1/3$
- $P(\text{Blue}|B) = 7/15$
- $P(\text{Blue}|C) = 2/15$

Calculation

Using Bayes' Theorem,

$$P(C|\text{Blue}) = \frac{P(\text{Blue}|C) \times P(C)}{P(\text{Blue})}$$

$$= \frac{P(\text{Blue}|C) \times P(C)}{P(\text{Blue})}$$

$$= \frac{(2 \times 1/4) + (7 \times 1/2) + (4 \times 1/4)}{(2 \times 1/4) + (7 \times 1/2) + (4 \times 1/4)} = \frac{2}{21}$$

$$P(\text{Blue}) = \frac{(2 \times 1/4) + (7 \times 1/2) + (4 \times 1/4)}{(2 \times 1/4) + (7 \times 1/2) + (4 \times 1/4)} = \frac{2}{21}$$

$$P(C|\text{Blue}) = \frac{2 \times 1/4}{2} = \frac{1}{4} = 0.0952$$

Answer

Therefore, the probability that the selected box was **Box C** is

$$\frac{2}{2+1+1} = 0.0952$$

$$= 9.52\%$$

$$= 0.0952 = 9.52\%$$

Conclusion: There is only a **9.52%** chance that the blue ball came from **Box C** because Box B has a higher probability of being selected and contains more blue balls.

2. A parking facility has three sections containing cars of different colors.

- Section A has 5 red, 3 blue, and 2 black cars.
- Section B has 4 red, 6 blue, and 5 black cars.
- Section C has 7 red, 2 blue, and 6 black cars.

A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage.

A randomly selected car from the chosen section turns out to be blue.

What is the probability that the car came from Section C?

Answer

Given:

Section A: 5 Red, 3 Blue, 2 Black (Total = 10)

Section B: 4 Red, 6 Blue, 5 Black (Total = 15)

Section C: 7 Red, 2 Blue, 6 Black (Total = 15)

Selection probabilities:

- $P(A) = \frac{1}{4}$
- $P(B) = \frac{2}{4}$
- $P(C) = \frac{1}{4}$

Probability of selecting a blue car:

- $P(\text{Blue}|A)=\frac{310}{103}$
- $P(\text{Blue}|B)=\frac{615}{252}$
- $P(\text{Blue}|C)=\frac{215}{152}$

Using Bayes' Theorem

$$P(C|\text{Blue})=P(\text{Blue}|C) \times P(C) / P(\text{Blue})$$

$$P(C|\text{Blue})=P(\text{Blue}|C) \times P(C) / P(\text{Blue})$$

First,

$$P(\text{Blue})=(310 \times 14) + (25 \times 12) + (215 \times 14)$$

$$P(\text{Blue})=(103 \times 41) + (52 \times 21) + (152 \times 41)$$

$$=340+15+130$$

$$=37120$$

$$=403+51+301=12037$$

Now,

$$P(C|\text{Blue})=215 \times 14 / 37120 = 437$$

$$P(C|\text{Blue})=12037 / 152 \times 41 = 374$$

Answer:

$$P(\text{Section C} | \text{Blue}) = 437 / 12037 \approx 0.1081$$

$$P(\text{Section C} | \text{Blue}) = 374 / 12037 \approx 0.1081$$

Therefore, the probability that the blue car came from Section C is 10.81%.

3. In a real-world medical image classification task for detecting breast cancer, a deep learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign).

The dataset consists of 1800 samples per class, with 20% of the data reserved for testing.

During evaluation, the model reports:

- **210 True Positives (TP)**
- **190 True Negatives (TN)**
- **30 False Positives (FP)**

Based on the given information:

- **Determine the missing False Negatives (FN) value.**
- **Calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier.**
- **Explain its effectiveness in medical diagnosis.**

Answer

Given:

- Samples per class = 1800
- Total samples = 3600
- Test data = 20%

Testing samples:

$$3600 \times 20\% = 720$$

Given:

- $TP = 210$
- $TN = 190$
- $FP = 30$
- $FN = ?$

Step 1: Calculate False Negative (FN)

$$TP + TN + FP + FN$$

$$= 720$$

$$= 210 + 190 + 30 + FN$$

$$= 210 + 190 + 30 + FN$$

$$= 430 + FN$$

$$= 430 + FN$$

$$= 290$$

Step 2: Accuracy

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}}$$

$$= \frac{210 + 190}{720} = 55.56\%$$

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$$\text{Accuracy} = \frac{TP + TN}{\text{Total}}$$

$$= \frac{210 + 190}{720}$$

$$= 55.56\%$$

Step 3: Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{210}{240} = 87.50\%$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{210}{240} = 87.50\%$$

Step 4: Sensitivity (Recall)

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$= \frac{210}{500} = 42\%$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$= \frac{210}{500} = 42\%$$

Step 5: Specificity

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$= \frac{190}{220} = 86.36\%$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$= \frac{190}{220} = 86.36\%$$

Answer:

- False Negative (FN) = 290
- Accuracy = 55.56%
- Precision = 87.50%
- Sensitivity (Recall) = 42%
- Specificity = 86.36%

Effectiveness:

The classifier has high precision and specificity, meaning it correctly identifies most predicted positive cases and negative cases. However, its sensitivity is low, indicating

that many actual cancer cases are missed. For medical diagnosis, improving sensitivity is important to reduce missed detections.

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs. (5 Marks)

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4

(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2

(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0

(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1

(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3

(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6

(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9

(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.

ii) Suggest two effective techniques to reduce overfitting in this model.

iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Answer

i) Identify the epoch at which the model begins to overfit.

The validation loss decreases from Epoch 1 to Epoch 3, but starts increasing from Epoch 4, while the training loss continues to decrease.

Answer: The model begins to overfit at Epoch 4.

ii) Suggest two effective techniques to reduce overfitting.

1. Early Stopping – Stop training when the validation loss starts increasing.
2. Dropout or L2 Regularization – Reduce model complexity and prevent memorization of training data.

iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Overfitting causes the model to learn the training data too closely, including noise and unnecessary details. As a result, it performs very well on training data but poorly on unseen or new data. This reduces the model's generalization ability and leads to lower prediction accuracy in real-world applications.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs. (5 Marks)

(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

(c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%

(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%

(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%

(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

(h) Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%

i) Identify the epoch at which the model starts overfitting.

ii) Explain why validation accuracy decreases despite training accuracy increasing.

iii) Suggest two methods to improve validation performance.

Answer

Concept

Training accuracy indicates how well the model performs on the training dataset, while validation accuracy measures its performance on unseen data. Comparing both values helps detect overfitting.

i) Identify the epoch at which the model starts overfitting.

Validation accuracy improves until **Epoch 5** and decreases from **Epoch 6**, while training accuracy continues to increase.

Answer: The model starts **overfitting at Epoch 6**.

ii) Explain why validation accuracy decreases despite training accuracy increasing.

The model starts memorizing the training data instead of learning general patterns. Therefore, it performs well on the training dataset but fails to perform equally well on unseen validation data.

iii) Suggest two methods to improve validation performance.

- **Early Stopping**
- **Data Augmentation or Dropout**

Conclusion

The model starts overfitting at **Epoch 6**. Applying early stopping and regularization techniques helps improve validation accuracy and the model's generalization ability.